

DAY – 12

AI Model Optimization

In this task, I focused on improving the performance and efficiency of the Virtual Try-On system by optimizing AI models used for pose detection, landmark extraction, segmentation, and try-on processing. Since Virtual Try-On systems depend heavily on body landmark accuracy and image processing speed, optimization plays an important role in improving fitting quality and user experience.

Landmark Detection Accuracy

Landmark detection is one of the most important components in a Virtual Try-On system. It helps identify body keypoints such as shoulders, elbows, hips, knees, and torso position. These landmarks are used to align clothing properly on a person.

During optimization, I explored ways to improve pose estimation accuracy so that garments can fit better according to body posture and body proportions.

Areas explored:

- Better shoulder and torso detection
- Improved body keypoint accuracy
- Reducing landmark misalignment
- Improving cloth positioning using pose estimation



Another important optimization area is **processing speed**. A Virtual Try-On system should ideally produce results quickly while maintaining fitting quality. Slow processing creates delays and reduces user experience.

To improve performance, lightweight models and optimized processing methods were explored.

Optimization ideas studied:

- Reducing unnecessary image processing steps
- Using lightweight AI models
- Faster pose estimation techniques
- Efficient resizing and masking methods

Recommendation Logic Improvement

Recommendation logic can improve the personalization of a Virtual Try-On system by suggesting clothing styles based on user measurements, body type, or previous selections.

- Basic optimization ideas explored include:
- Clothing suggestions based on body proportions
- Better fitting recommendations
- Category-based garment suggestions
- Faster recommendation processing

Conclusion

Through this task, I explored different AI optimization techniques used in Virtual Try-On systems and studied how landmark detection, segmentation, inference speed, and recommendation logic affect system performance. I also compared different AI models and understood how lightweight and optimized approaches can improve real-time try-on quality while maintaining efficiency.